DISK-TECTIVE WORK: THE LINK BETWEEN INNER DISK GAS AND TERRESTRIAL PLANET FORMATION

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Theo Olsen[a], Annie Dickson-Vandervelde[a]
[a]Vassar College

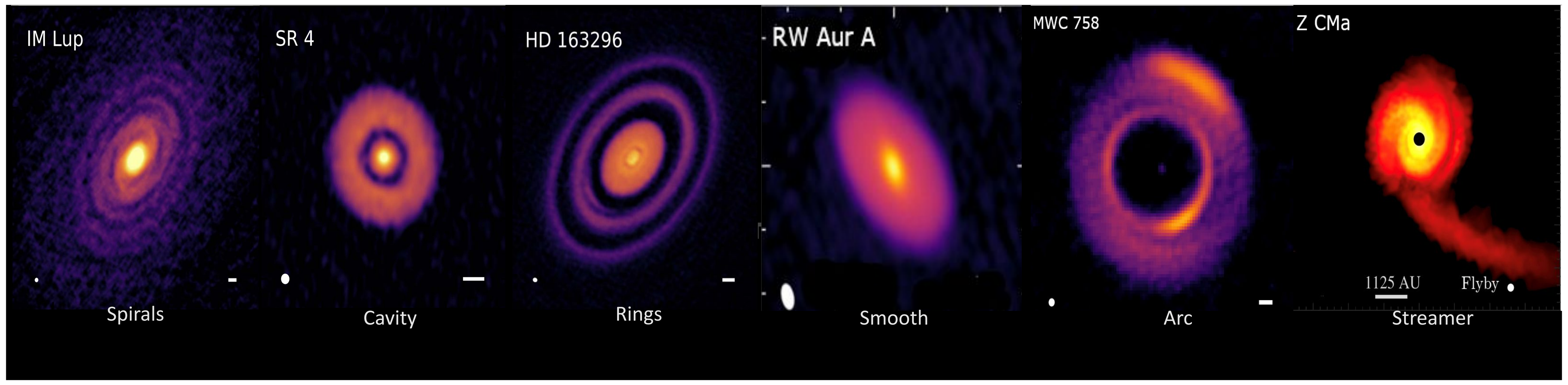
INTRO

The inner regions of protoplanetary disks are thought to influence how planets form, but the exact effects remain unclear. By studying these disks, we learn how their features and chemistry impact the development of planets and what conditions may lead to habitable worlds.

What We Know So Far:

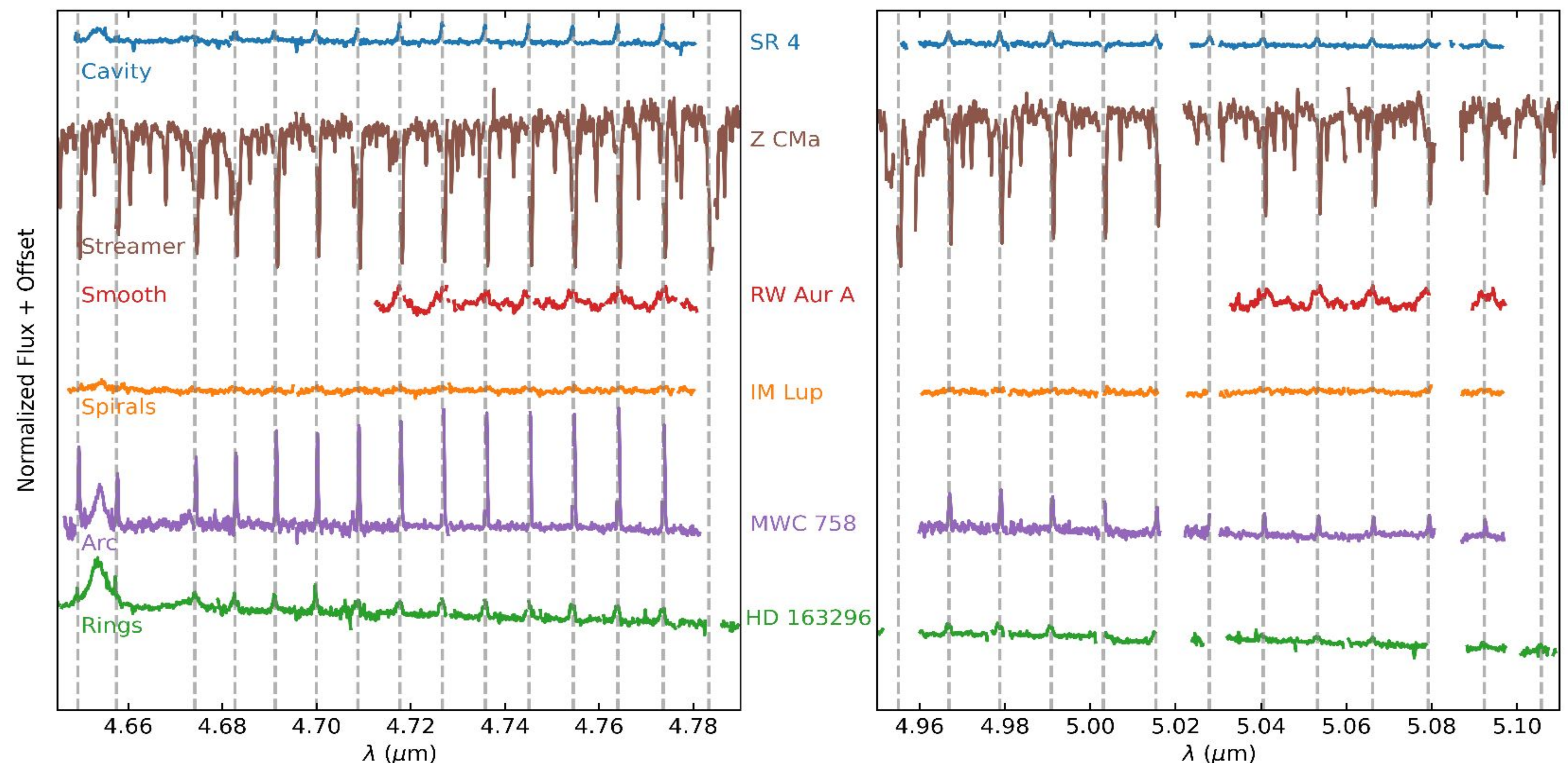
- Carbon monoxide (CO) traces the hot gas in the inner disk, revealing dynamics and temperatures in key planet-forming areas.
- The inner disk, where terrestrial planets like Earth form, interacts closely with the star through processes like accretion.
- Studies, like those from the DSHARP team[1], have revealed ring structures, spirals, and other dust substructures in disks, offering clues to planetary formation.

Our Research Focus: We analyze 186 protoplanetary disks using CO emissions and high-resolution images from ALMA to explore how inner disk chemistry connects to outer disk structures. Understanding these links could unlock the secrets of how planets form and evolve in their early stages.



Continuum emission images of: IM Lup[1] SR 4[1] HD 163296[1] RW Aur A[2] MWC 758[3] Z CMa[4]

SPECTRA



METHODS

Our study analyzes 186 stars with protoplanetary disks, including low-mass T Tauri stars and high-mass Herbig Ae/Be stars, spanning different spectral types and disk classifications (Class I, II, III, and transition disks).

Data Collection:

- Keck-NIRSPEC Spectra: We used the high-resolution infrared spectrometer on the Keck II telescope, collecting observations from 2001-2019. We analyzed CO emission lines to measure key parameters, including:
  - Line fluxes: Strength of CO emission at expected wavelengths.
  - Continuum levels: Background radiation from each star, modeled using blackbody curves.
  - Line widths: Determined by Doppler shifts to gauge disk dynamics.
  - Retrieved Properties: Using slab modeling, we calculated column densities, temperatures, and emitting areas for disks with sufficient CO emission.
- ALMA Disk Images: We analyzed 69 disks imaged at millimeter wavelengths. Disks were sorted into six categories (smooth, cavity, rings, spirals, arc, and streamer) based on substructure. We focused on large-scale dust substructures and excluded edge-on or shadowed disks.

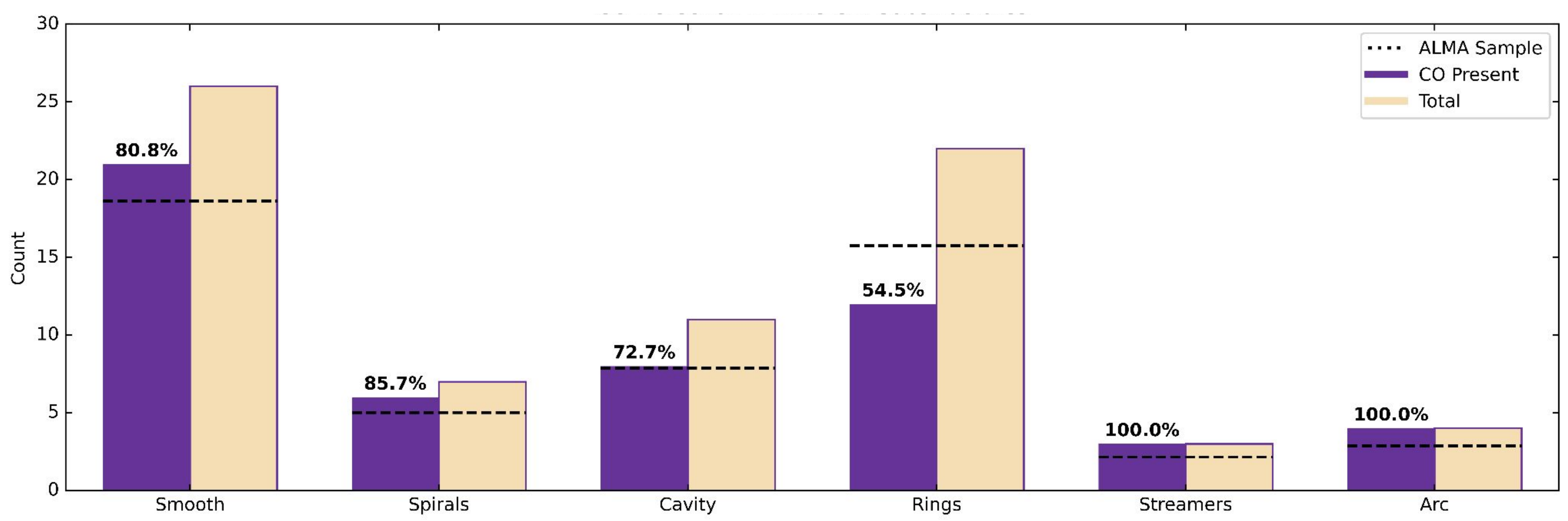
Our approach links inner disk CO properties to large-scale dust features, testing how dynamics and chemistry in the inner disk correlate with external disk morphology.

FUN FACTS

> Birthplaces of Planets: Protoplanetary disks are massive clouds of gas and dust around young stars. Over time, this material clumps together to form planets, moons, and other celestial bodies.

> Where Earth-Like Planets Form: The inner regions of these disks (within 5 AU of the star) are where rocky, terrestrial planets—like Earth—are thought to form[5].

CO Detection Rate Across Substructures



This plot demonstrates the CO (1-0) detection rates across our ALMA imaged sample, categorized by substructure. The yellow bars indicate the total number of disks with a given substructure, while the purple bars indicate the disks of that substructure with CO(1-0) detection. The dotted line shows where the purple bar would have to be to match the detection rate of the entire sample.

RESULTS

Stellar Properties vs. CO Detection:

- No correlation was found between stellar type and CO detection. Both T Tauri stars (74% detection rate) and Herbig Ae/Be stars (70%) showed similar rates, despite a smaller sample of Herbig stars.
- Substructures, such as rings and cavities, were present across all stellar types, with the majority being K and M-type stars.

CO Emissions vs. Substructure:

- Among the 69 disks with imaged substructures, most were either smooth or ringed.
- Disks with rings showed a lower CO detection rate compared to the full sample, while smooth and cavity disks had similar detection rates (within 10% of the overall average).
- Substructures like spirals, arcs, and streamers were less common and not the focus of this study due to limited sample sizes.

CO Detection vs. Emission Properties:

- CO (2-1) and ^{13}CO showed similar detection rates to the standard CO (1-0), with no significant differences across substructures.
- Smooth disks had a lower median emitting area, likely because 90% of them were compact disks (dust radius <50 AU), while ringed disks were generally more extended.
- Smooth disks had higher column densities, likely due to the absence of gaps.
- Temperature distributions were similar between smooth and ringed disks.
- Higher star-disk system mass was associated with a greater likelihood of disk substructure, indicating that more massive systems tend to develop more complex disk features.

SUMMARY

- We reviewed and analyzed Keck-NIRSPEC spectra, calculating line fluxes, continuums, and linewidths for CO (1-0), CO (2-1), and ^{13}CO emissions for all targets.
- Comparing substructure classifications with CO spectroscopy, we found that ringed disks exhibit a lower CO (1-0) detection rate in their inner regions compared to disks without substructures.
- Additional trends between substructures and characteristics like system mass, column density, and dust radius were identified, though further research is needed to confirm these findings.

This analysis provides new insights into how disk structure influences inner disk chemistry and planet-forming conditions.

CITATIONS

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